

Genetic report for longevity

Caffeine and Stimulants

Caffeine metabolism (CYP1A2, AHR), sleep and anxiety sensitivity (ADORA2A), consumption of coffee and tea, nicotine and other stimulants

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Caffeine is the most used stimulant and one of the most studied in genetics: CYP1A2 eliminates about 95% of it, AHR regulates CYP1A2 and A2A adenosine receptor (ADORA2A) decides how much it moves in sleep and anxiety. This report reads these genes, the loci of coffee and tea consumption, four polygenic consumption scores, a caffeine metabolization score, and the genetics of nicotine and other stimulants (modafinil, methylphenidate). The genetic plan already limits caffeine to 200 mg before midday; here we see why and what weighs the most in this decision.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

20 variants	19 in the chip	0 imputed (DR2≥0.8)	1 not found	7 Scores with percentile
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Overall reading. The relationship with caffeine is favourable to typical. In elimination: CYP1A2 *1F A/C (a copy of the inducible, fast allele) and CYP1A1-CYP1A2 rs2472297 C/C (without the fast elimination allele); the 3 loci score of metabolism is at the 82nd percentile (informative). In the set, normal elimination without slow metabolizer signal. In sensitivity: ADORA2A C/C, the genotype less associated with anxiety and insomnia with caffeine; Health Report puts caffeine sensitivity at the 18th percentile and caffeine sleep problems in the 13 ("less likely"). The plan uses the tag "higher" of another report; the limits of the plan are as they are. In consumption, the scores are divided: drinking coffee at the 83rd percentile (PGS of 48 variants), ground coffee in the 65, consumption of coffee in the 44 and score of 66 loci in the 32; tea in the 63. All informative. In nicotine: PGS of having smoked at percentile 1.6 (favourable), CYP2A6 without alleles *2 and *9 and CHRNA5 G/A (1 copy of the dependency allele, typical). COMT G/G (Val/Val) and ADA C/C without judgment.

Favourable

- **Have smoked (974 variants):** the 2nd percentile
- **Sleep problems related to caffeine (2 variant score):** the 13th percentile (Health Report)
- **ADORA2A rs5751876 CC:** ADORA2A 1976C>T: T allele is associated with more anxiety and worse sleep with caffeine; C/C is the least sensitive genotype.

Unfavourable

None.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 19 chip genotypes, 0 needed complement, 2 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Metabolism of caffeine

CYP1A2 (*1F, rs762551; and the CYP1A1-CYP1A2, rs2472297), AHR (which induces CYP1A2) and POR (the CYP electron donor). The half-life of caffeine varies from 2 to 10 hours between people; tobacco accelerates it and some drugs stop it.

Ageing mechanisms: Deregulation of Nutritional Sensing; Vascular Aging

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Metabolization of caffeine (paraxanthine/caffeine ratio)	No judgement (informative)	the 89th percentile · z=1.34	Higher score : faster elimination. Information. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 3 of 3 loci with genotype. Studies: Cornelis MC (Caffeine metabolite (plasma 1.7-dimethylxanthine (paraxanthine) to 1.3,7-trimethylxanthine (caffeine) ratio)). Variants with genotype: 3 of 3 (covering of 100.0% coefficients).

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (5)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP1A2 (*1F)	rs762551 15:74749576 · C/A/G	AC	chip	—	moderate	CYP1A2 -163C>A: allele A makes CYP1A2 more inducible (fast elimination, especially in smokers); C/C and A/C are slower. A/C: an *1F allele (inductive); intermediate elimination in a non-smoker.
CYP1A1-CYP1A2	rs2472297 15:74735539 · C/A/T	CC	chip	T (0; 22%)	strong	Shared promoter CYP1A1/CYP1A2 signal: T allele accelerates elimination and is associated with more coffee per day. C/C: without the fast elimination T allele (the plan uses this data to limit caffeine). Information. Meaning: GWAS Catalog: Coffee consumption (Zhong VW 2019; p=9e-157); T ↑ allele
AHR	rs4410790 7:17244953 · T/C	CC	chip	C (2; 62%)	strong	AHR: induces CYP1A2; strong locus of coffee consumption. More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: Caffeine consumption from coffee or tea (Said MA 2020; p=2e-249); T ↓ allele
AHR	rs6968865 7:17247645 · A/T	TT	chip	T (2; 61%)	moderate	AHR: second sign of caffeine consumption and metabolism. More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: F-chocolate/coffee like (derived food-like factor) (May-Wilson S 2022; p=2e-25); T ↑ allele
POR	rs17685 7:75986787 · G/A/C	GA	chip	A (1; 29%)	moderate	POR: oxidoreductase of the cytochrome P450; accelerates CYP. More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: Coffee consumption (cups per day) (Kennedy OJ 2019; p=8e-20); allele A ↑

Sensitivity: sleep, anxiety and heart

ADORA2A (adenosine A2A receptor) is the main gene of insomnia and anxiety induced by caffeine; ADA affects sleep pressure (adenosine). The Circadian and Sleep Rhythm report has already read these genes.

Ageing mechanisms: Sleep disturbance; Circadian disorders

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Caffeine sensitivity (3 variant score)	No judgement (informative)	the 18th percentile	It labels "lower" in the Health Report; the plan uses the tag "higher" of another report. Source: Health Report.
Sleep problems related to caffeine (2 variant score)	Favourable	the 13th percentile	It labels "less likely". Source: Health Report.

Favourable

- Sleep problems linked to caffeine (2 variant score) (the 13th percentile)
- ADORA2A rs5751876 CC

Unfavourable

None.

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADORA2A	rs5751876 22:24441333 · T/C	CC	chip	T (0; 39%)	moderate	ADORA2A 1976C>T: T allele is associated with more anxiety and worse sleep with caffeine; C/C is the least sensitive genotype.

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ADA	rs73598374 20:44651586 · C/A/G/T	CC	chip	—	moderate	ADA G22N: A allele lowers the activity of adenosine deaminase and deepens sleep; informative.

Coffee and tea consumption

The genes that accelerate the elimination lead to more coffee; other loci (GCKR, MLXIPL, BDNF, ABCG2) affect the reward and metabolism. Consumption scores measure the trend, not the ideal dose.

Ageing mechanisms: Nutrient Sensing Deregulation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Coffee consumption (3154 variants)	No judgement (informative)	the 44th percentile · z=-0.14	Higher score = more coffee per day. Information. PGS001126 (PGS Catalog; snpnet; Coffee intake; PGP000244, PMID 35324888). Variants with genotype: 2762 of 3154 (covering of 88.2% coefficients).
Grounded coffee (9694 variants)	No judgement (informative)	the 65th percentile · z=0.42	Higher score = more ground coffee. Information. PGS001124 (PGS Catalog; snpnet; Coffee consumption; PGP000244, PMID 35324888). Variants with genotype: 8493 of 9694 (covering of 88.6% coefficients).
Coffee drink (48 variants)	No judgement (informative)	the 83rd percentile · z=0.97	Higher score = more likely to drink coffee. Information. PGS001123 (PGS Catalog; snpnet; Coffee consumption; PGP000244, PMID 35324888). Variants with genotype: 35 of 48 (covering of 69.4% coefficients).
Tea consumption (3119 variants)	No judgement (informative)	the 63rd percentile · z=0.33	Higher score = more tea. Information. PGS000994 (PGS Catalog; snpnet; Tea intake; PGP000244, PMID 35324888). Variants with genotype: 2768 of 3119 (covering of 89.2% coefficients).
Coffee consumption (GWAS loci)	No judgement (informative)	the 32nd percentile · z=-0.45	Higher score = more coffee. Information. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 56 of 66 loci with genotype. Studies: Amin N (Coffee consumption); Cole JB (Coffee consumption) Cornelis MC (Coffee consumption) Jia H (Coffee consumption); Kennedy OJ (Coffee consumption) Nakagawa-Senda H (Coffee consumption); Said MA (Caffeine consumption from coffee or tea); Said MA (Caffeine consumption from coffee). Variants with genotype: 56 of 66 (covering of 74.9% coefficients).

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (7)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GCKR	rs1260326 · 2:27508073 · T/C/G	TC	chip	C (1; 59%)	moderate	GCKR P446L: locus of coffee consumption (and triglycerides). More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: Coffee consumption (Zhong VW 2019; p=1e-19); C ↑ allele
MLXIPL	rs7800944 · 7:73621527 · T/C	TT	chip	C (0; 26%)	moderate	MLXIPL (ChREBP): locus of coffee consumption. More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: Coffee consumption (cups per day) (Cornelis MC 2014; p=3e-11); T ↓ allele
BDNF	rs6265 · 11:27658369 · C/T	TC	chip	C (1; 80%)	moderate	BDNF Val66Met: locus of coffee consumption (reward). More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: Caffeine consumption from coffee or tea (Said MA 2020; p=3e-13); allele C ↑
ABCG2	rs1481012 · 4:88117930 · A/G	AA	chip	A (2; 91%)	moderate	ABCG2: transporter; locus of coffee consumption. More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: Coffee consumption

						(cups per day) (Kennedy OJ 2019; p=2e-12); G ↓ allele
EFCAB5	rs9902453 17:30022077 · G/A	AA	chip	G (0; 43%)	moderate	EFCAB5: locus of coffee consumption. More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: Coffee consumption (cups per day) (Cornelis MC 2014; p=3e-08); Allele A ↓
NRCAM	rs382140 7:108141755 · A/G/T	GG	chip	A (0; 19%)	moderate	NRCAM: neuronal adhesion; coffee consumption. More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: Coffee consumption (Amin N 2011; p=4e-09); Allele A ↑
SEC16B	rs574367 1:177904075 · G/T	GG	chip	T (0; 19%)	moderate	SEC16B: coffee consumption. More coffee or faster elimination is not, by itself, favourable or unfavourable: informative. Meaning: GWAS Catalog: Coffee consumption (Zhong VW 2019; p=5e-10); T ↑ allele

Nicotine and other stimulants

CYP2A6 (nicotine elimination), CHRNA5 (dependence), CES1 (methylphenidate), COMT (modafinil response). The most cited dopamine genes (DRD4, SLC6A3) are VNTR and do not read by chip.

Ageing mechanisms: Intercellular communication alteration; Oxidative stress

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Have smoked (974 variants)	Favourable	the 2nd percentile · z=-2.31	Higher score : more tendency to smoke. PGS001129 (PGS Catalog; snpnet; Smoking status (ever vs never smokers); PGP000244, PMID 35324888). Variants with genotype: 833 of 974 (covering of 86.6% coefficients).

Favourable

- Have smoked (974 variants) (the 2nd percentile)

Unfavourable

None.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CHRNA5	rs16969968 · 15:78590583 · G/A	GA	chip	A (1; 37%)	strong	CHRNA5 D398N: allele A increases nicotine dependence and the number of cigarettes.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP2A6 (*2)	rs1801272 · 19:40848628 · A/T	AA	chip	—	strong	CYP2A6*2 (L160H): null allele; slower elimination of nicotine. Informative (do not smoke). A/A: common genotype, without the *2 allele.
CYP2A6 (*9)	rs28399433 · 19:40850474 · A/C/G/T	AA	chip	—	moderate	CYP2A6*9: less expression; slower elimination of nicotine. Information. A/A: without the *9 allele.
COMT	rs4680 · 22:19963748 · G/A	GG	chip	—	weak	COMT Val158Met Val/Val was associated with more cognitive benefit of modafinil in small studies. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CES1	rs71647871 · 16:55823658 · C/A/T	—	SNP not found in the data (Imputed with DR2=0.54 < 0.8)	A (—; —)	moderate	CES1 G143E: loss of function allele; much higher methylphenidate levels. I do not deal with quality in this chip; the pharmacogenetic profile indicates CC (without G143E).

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Intercellular Communication	—	Have smoked (974 variants) (the 2nd percentile)	CHRNA5 GA
Nutrient Sensing Deregulation	—	—	—
Vascular Aging	—	—	—
Sleep disturbance	—	Sleep problems linked to caffeine (2 variant score) (the 13th percentile); ADORA2A CC	—
Circadian disorders	—	Sleep problems linked to caffeine (2 variant score) (the 13th percentile); ADORA2A CC	—
Oxidative stress	—	Have smoked (974 variants) (the 2nd percentile)	CHRNA5 GA

Practical part

Ideas for caffeine and stimulants, crossed with the genetic plan v2 and with the current medication (isotretinoin, retatrutide, dapagliflozin, telmisartan, tadalafil, rapamycin, lithium in low dose). The current stack remains as it is; everything is suggestion (not applied). With this genetic, caffeine is well tolerated; what counts more are drug interactions and time of use.

1. Maintain the limit of the plan: up to 200 mg, only in the morning

Genetics: ADORA2A C/C; p18 (Health Report) sensitivity

Even with favourable genetics, caffeine delays deep sleep for hours. Keep up to ~200 mg per day and the last dose before midday.

V2 genetic plan and medication: Already in the plan. • *suggestion (not applied)*

2. Prefer filtered coffee

Genetics: Typical to high coffee consumption

The paper filter retains the coffee and kahweol, which rise the LDL. The boiled coffee, of press or Turkish is not filtered.

V2 genetic plan and medication: Compatible with the lower (LDL) plan. • *suggestion (not applied)*

3. Caffeine in the morning and melatonin in the night

Genetics: CYP1A2 *1F A/C

Melatonin (0.3–1 mg, the dose of the plan) is also eliminated by CYP1A2. Late caffeine competes with it and prolongs its effect the following day; with caffeine only in the morning there is no overlap.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

4. Stable consumption due to lithium

Genetics: Normal elimination (CYP1A2)

Caffeine increases the elimination of lithium by the kidney. With the low dose the effect is small, but abrupt changes (stopping coffee suddenly or duplicating it) move at the level. Maintain a regular consumption.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

5. L-theanine with coffee (suggestion)

Genetics: COMT Val/Val; ADORA2A C/C

100–200 mg of L-theanine with caffeine smoothes the peak agitation in some small studies. Paraxanthine is an alternative to caffeine with less data. The two remain only as a suggestion.

V2 genetic plan and medication: It's not in the plan. • *suggestion (not applied)*

Avoiding or with caution

- Caffeine with ciprofloxacin or fluvoxamine: blocking CYP1A2 and multiplying the time of caffeine in blood. Reduce or stop these days. *suggestion (not applied)*
- Modafinil or armodafinil without medical follow-up: induce CYP3A4 and lower rapamycin levels. *suggestion (not applied)*
- Nicotine bags or tablets: the smoking score is low, but nicotine dependence (CHRNA5) does not depend only on the will. *suggestion (not applied)*
- Energy drinks with more than 200 mg or in the afternoon. *suggestion (not applied)*
- Coffee not filtered every day (sobe LDL). *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. VNTR of DRD4 and SLC6A3 and the deletions of CYP2A6 (*4) are not read.
- The consumption scores come from questionnaires (teaves per day); measure a trend of behavior, not tolerance or safe dose.
- CYP1A2 is highly induced by tobacco, grilled meat and crucifers, and blocked by some drugs; the phenotype can move away from the genotype.
- A score of cigarettes loci per day (219 loci of different studies, counting without weighting) gave an extreme value ($z = -4.4$), in the same direction as PGS of having smoked (percentile 1.6). For being extreme and mixing studies, it was not interpreted; PGS remains.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001126 (PGS Catalog; snpnet; Coffee intake; PGP000244, PMID 35324888 · PGS001124 (PGS Catalog; snpnet; Coffee consumption; PGP000244, PMID 35324888 · PGS001123 (PGS Catalog; snpnet; Coffee consumption; PGP000244, PMID 35324888 · PGS000994 (PGS Catalog; snpnet; Tea intake; PGP000244, PMID 35324888 · PGS001129 (PGS Catalog; snpnet; Smoking status (ever vs never smokers); PGP000244, PMID 35324888).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Cornelis MC et al. Genome-wide meta-analysis identifies six novel loci associated with habitual coffee consumption. *Mol Psychiatry* 2015.
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- Childs E et al. Association between ADORA2A and DRD2 polymorphisms and caffeine-induced anxiety. *Neuropsychopharmacology* 2008.
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- Zhu HJ et al. Two CES1 gene mutations lead to dysfunctional carboxylesterase 1 activity in man: clinical significance and molecular basis. *Am J Hum Genet* 2008.
- Tanigawa Y et al. Significant sparse polygenic risk scores across 813 traits in UK Biobank. *PLoS Genet* 2022 (PGS000994, PGS001123-PGS001129).
- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

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